

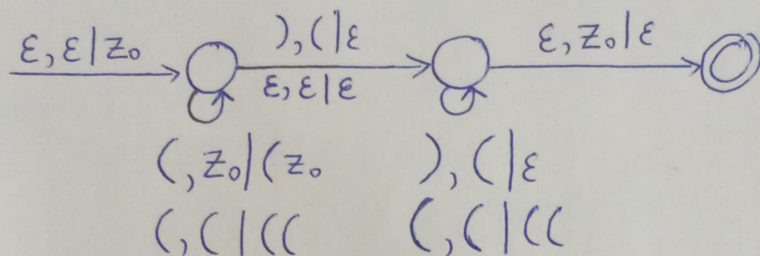
TOA - Assignment 3

Name : Sana Munir Alam

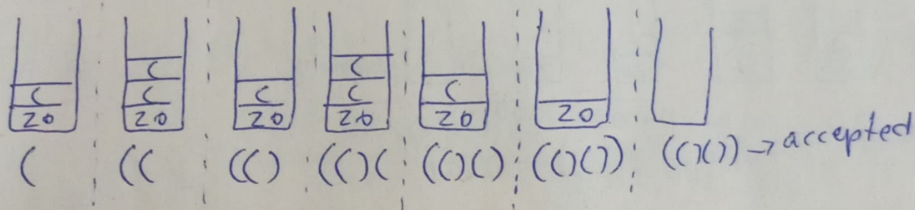
Roll #: 24K-0573

Class : BCS - 3E

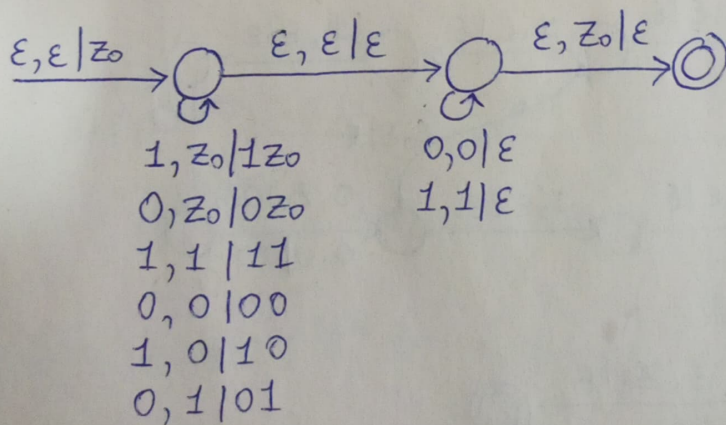
Q11) PDA of balanced parenthesis. $\Sigma = (,)$



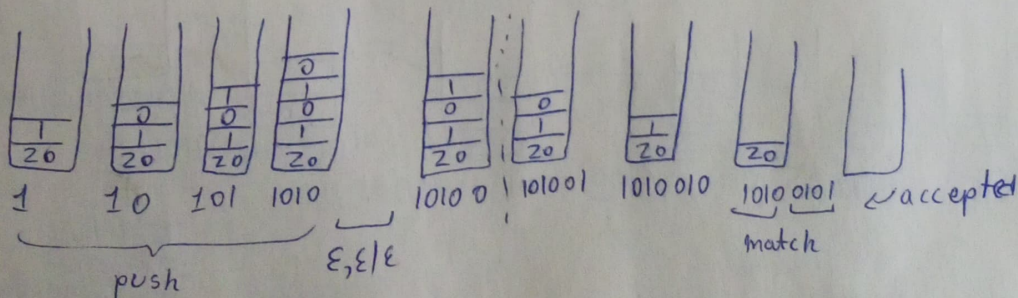
eg: $(()) ()$



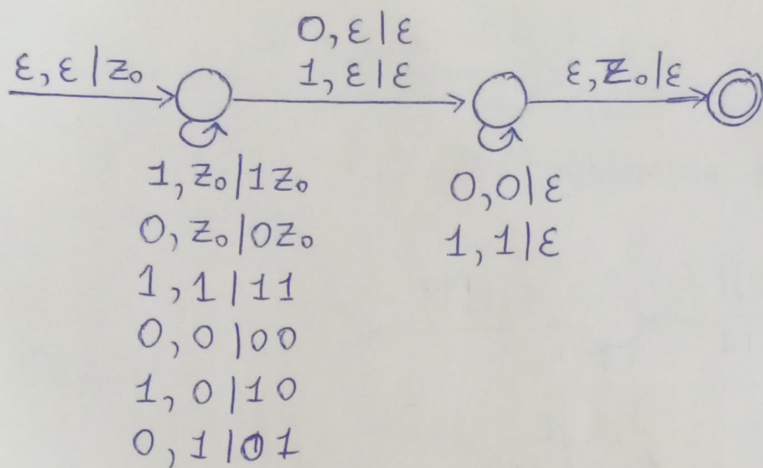
ii) Even Palindrome PDA



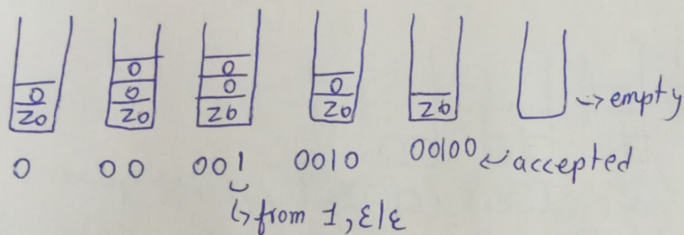
eg: 10100101



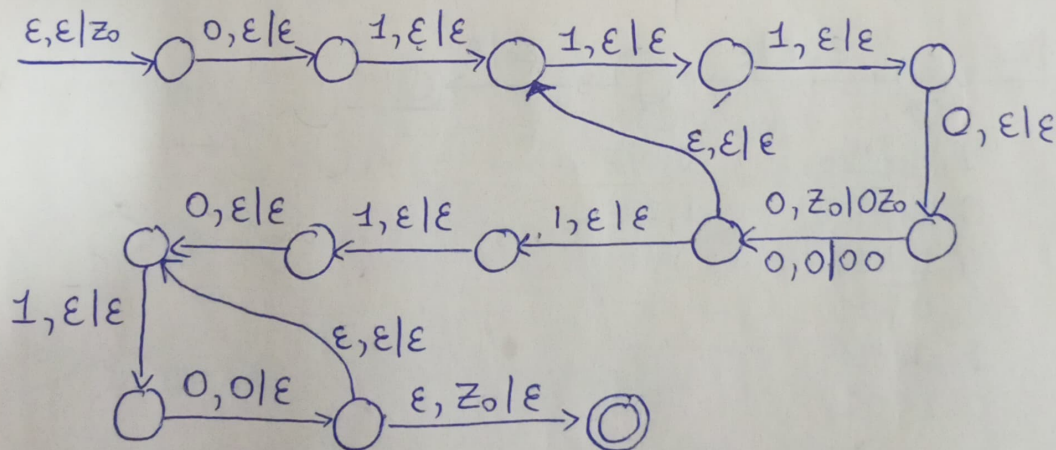
ii) Odd palindrome



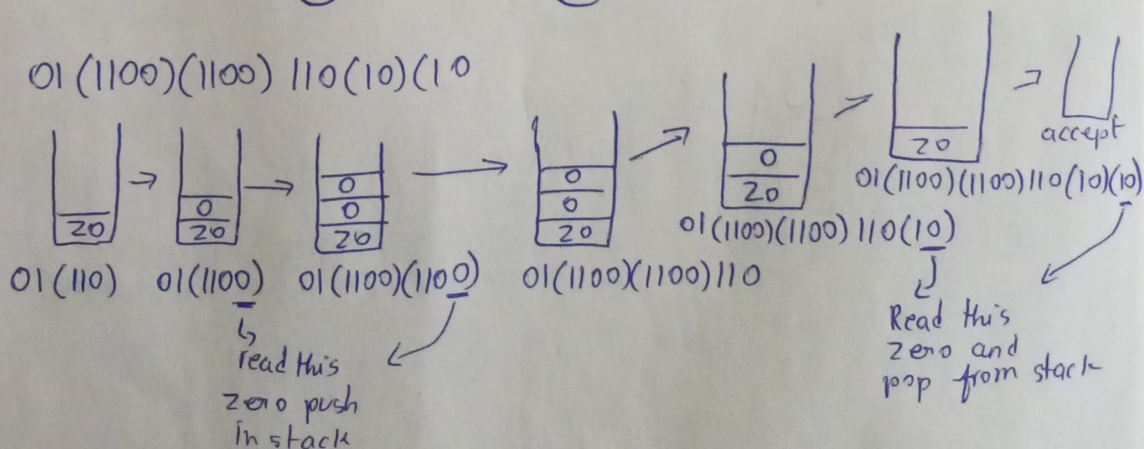
eg:- 00100



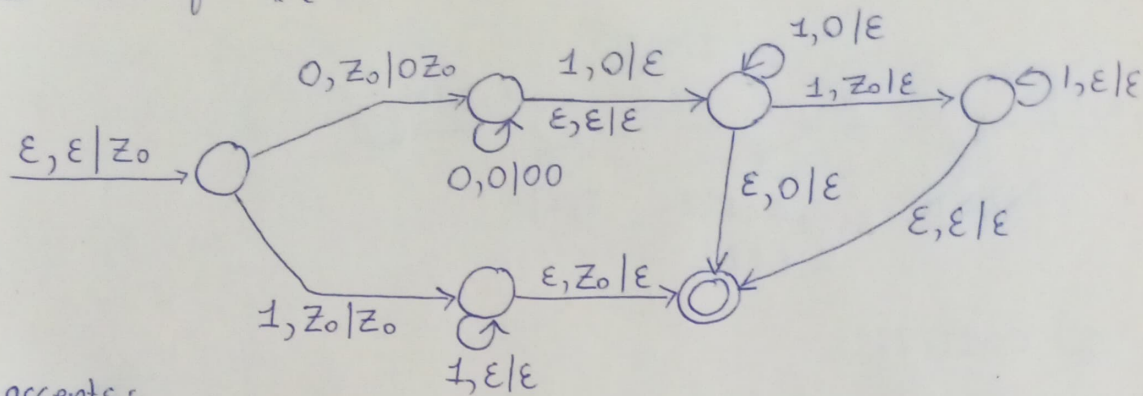
iii) PDA for language $\{01(1100)^n 110(10)^n \mid n \geq 0\}$



eg: 01(1100)(1100)110(10)(10)



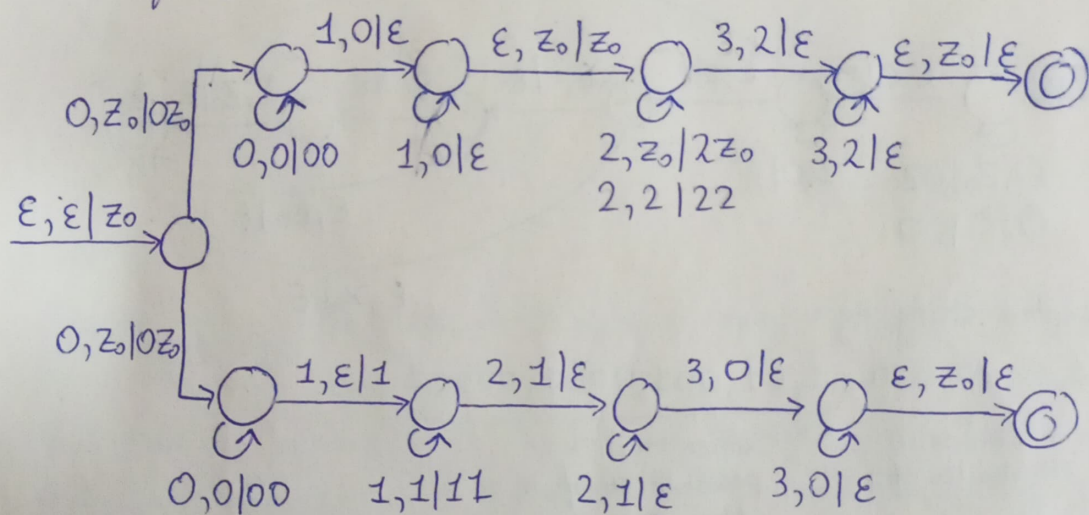
iv) PDA for $\{0^n 1^m \mid n \neq m\}$



accepts:

1, 001, 00111, 1111 etc

v) PDA for $\{(0^n 1^n 2^m 3^n \mid n \geq 1, m \geq 1) \text{ or } (0^n 1^m 2^m 3^n \mid n \geq 1, m \geq 1)\}$

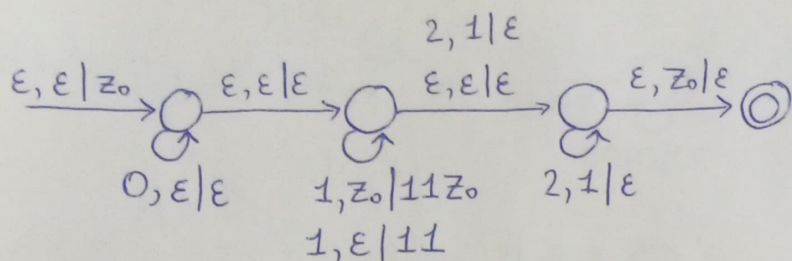


vi) PDA for $\{0^i 1^j 2^k \mid k \leq i, k \leq j\}$

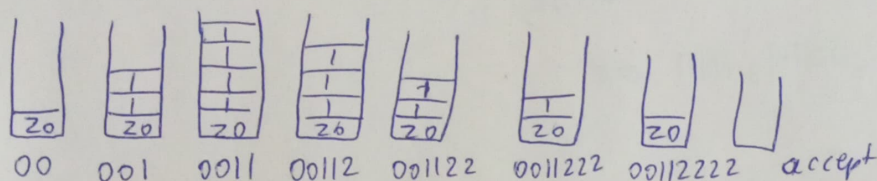
Not solvable with a single stack PDA.

vii) Not solvable with a single stack PPA.

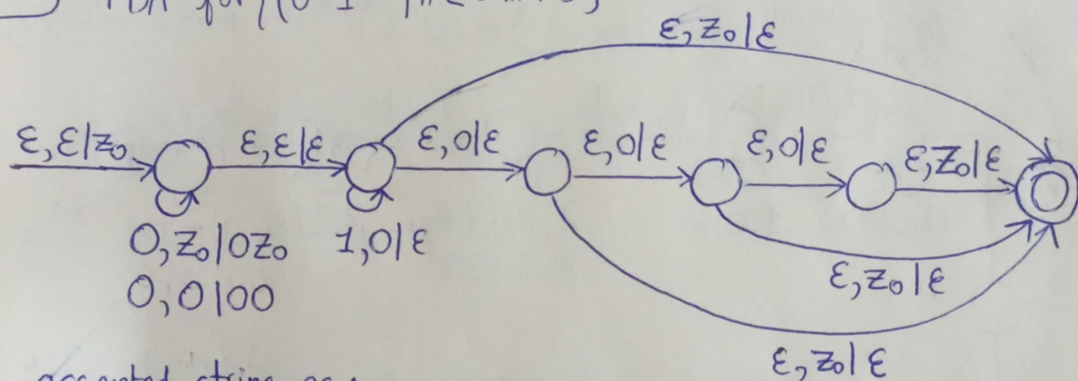
Viii) PDA for $\{0^n 1^m 2^{2m} \mid n, m \geq 0\}$



eg: 00112222



ix) PDA for $\{0^n 1^m \mid n \leq m+3\}$



accepted string eg,

$\epsilon, 0, 00, 000, 1, 01, 001, 0001, 00001, \dots$
 when $m=0$ possibilities of n when $m=1$ possibilities of n from 0 to 4

X) Not solvable with one stack.

Q2a) $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$

$Q = \{q_0, q_n, q_f\}$
 ↓ ↓ ↓
 initial main final

$\Sigma = \{ (,), \{, \}, [,] \}$

$\Gamma = \{ (, \{, [, Z_0 \}$

q_0 = start state

Z_0 = Initial stack symbol

$F = \{q_f\}$

Transition Function

	State	Input/Read	Pop	→ New State	Push
→	q_0	ϵZ_0	ϵ	q_0	ϵZ_0
	q_0	$\{$	ϵ	q_0	$\{$
	q_0	$($	ϵ	q_0	$($
	q_0	$[$	ϵ	q_0	$[$
	q_0	ϵ	ϵ	q_1	ϵ
	q_1	$)$	$($	q_1	ϵ
	q_1	$\}$	$\{$	q_1	ϵ
	q_1	$]$	$[$	q_1	ϵ
	q_1	$($	ϵ	q_1	$($
	q_1	$\{$	ϵ	q_1	$\{$
	q_1	$[$	ϵ	q_1	$[$
	q_1	ϵ	Z_0	q_f	ϵ
*	q_f	ϵ	ϵ	q_f	ϵ

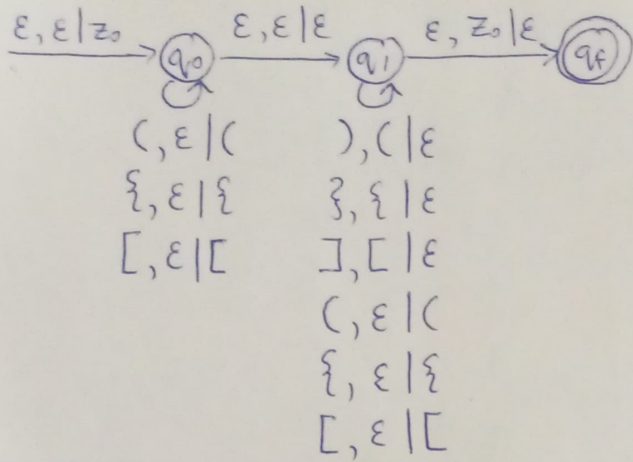
b) Working Principle

Push on opening:- When the PDA reads an opening symbol $(, \{, [$ it pushes the same symbol onto the stack. This records the expectation of a matching close later. The stack therefore holds the sequence of unclosed openings, with the most recent (innermost) opening on the top.

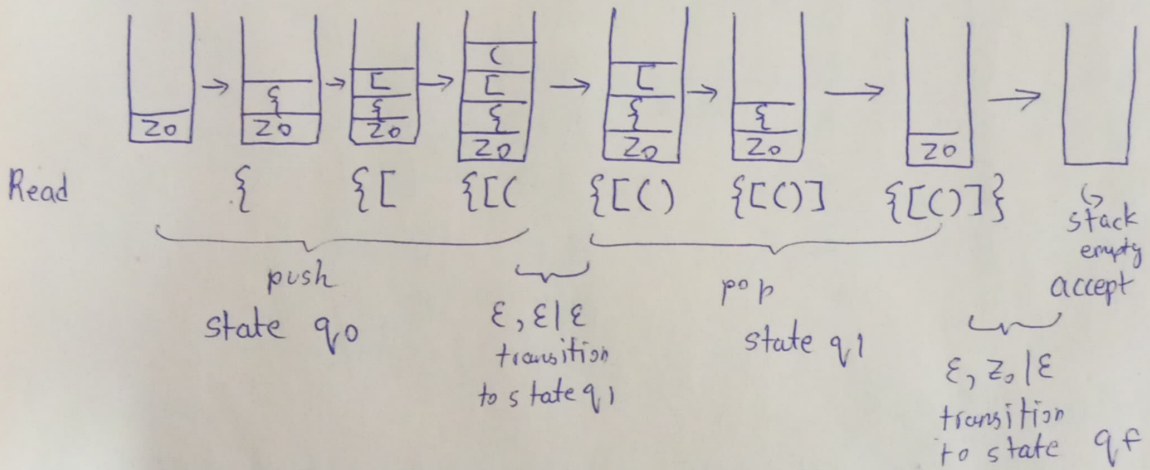
Pop on matching close:- When the PDA reads an ^{closing} ~~opening~~ symbol $), \}, \]$ it attempts to pop the stack only if the stack top is the matching opening symbol: $($ for $)$, $\{$ for $\}$, $[$ for $]$. If the top matches, the symbol is popped and processing continues. If it does not match there is no transition and that computation branch fails. This enforces both type-matching and correct nesting.

Final Acceptance:- After the input is fully read, the PDA checks that only the bottom marker Z_0 remains (i.e. all openings were closed). If so it does an ϵ -move to the accepting final state q_f . If unmatched opening remain, stack top will not be Z_0 and the machine can't reach q_f .

c) State Transition Diagram

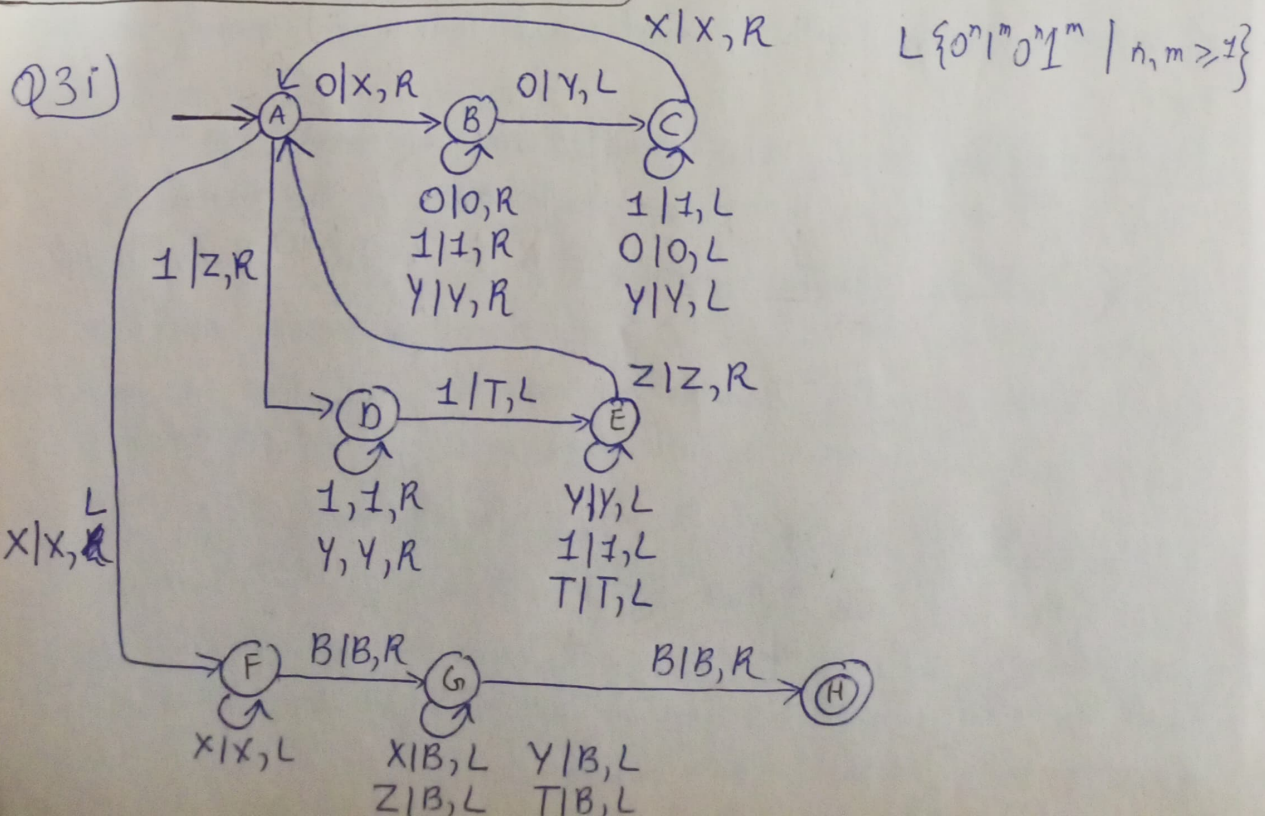


d) eg, Test string $\{ [()] \}$



e) CFG

$$S \rightarrow (S) \mid \{S\} \mid [S] \mid \epsilon$$

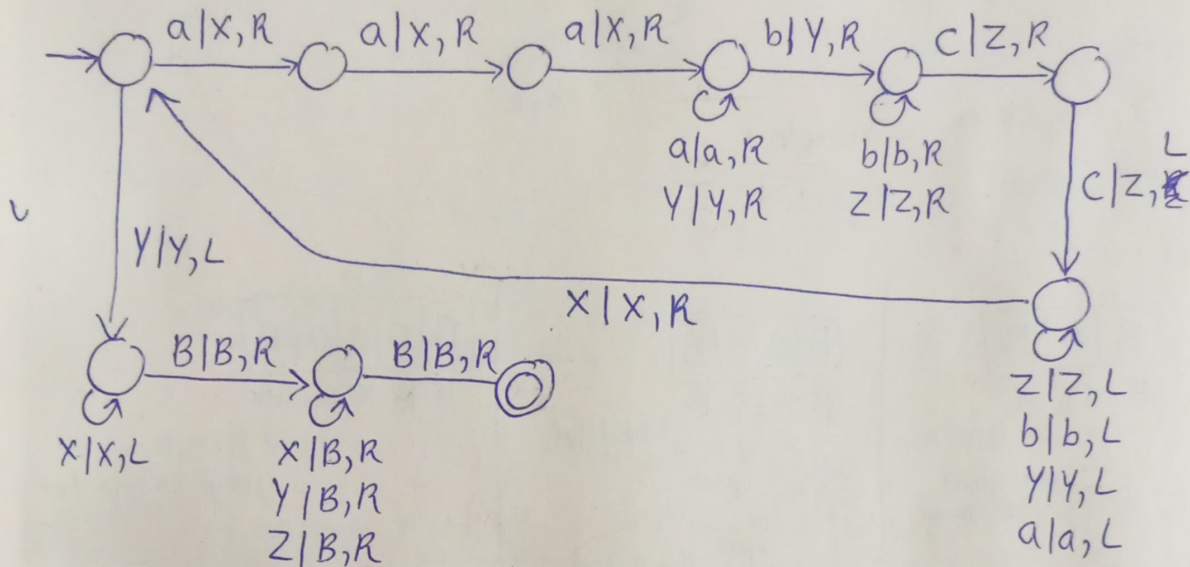


eg string: B 0001100011 B accept

X	Y	
XX	YY	
XXX	YYY	
XXXZ	YYY Z	
XXXZZ	YY Y TT	
BBBBB	BBBBB	accept

Rejected: 0, 1, 01, 010
10, 101 and so on

ii) $L_2 = \{a^{3n} b^n c^{2n} \mid n \geq 2\}$



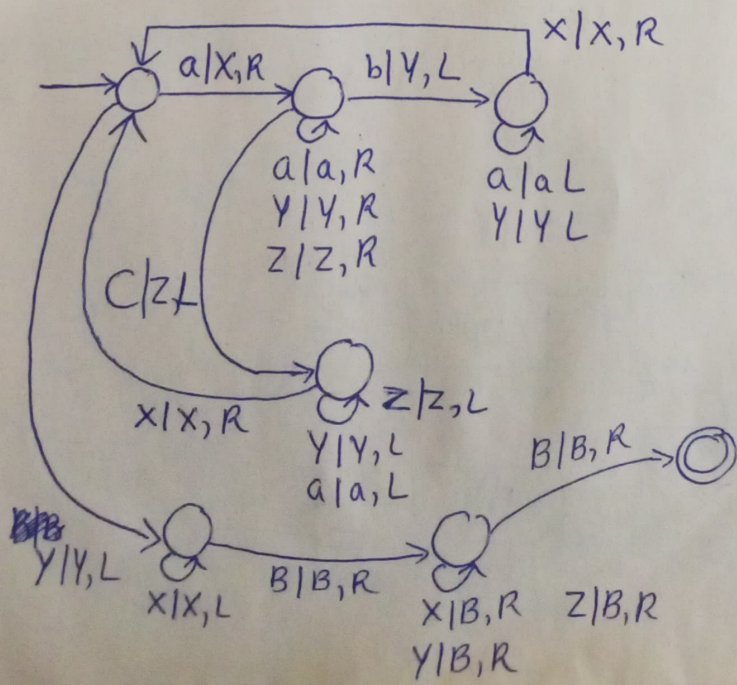
eg: string

B aaaaaa bb cccc B
 XXX Y ZZ
 XXX Y ZZ
 BBB BBB BB BBB accept

rejected string:

a, b, c, abc
 aaabbbcccccc

iii) $L = \{a^{n+m} b^n c^m \mid n \geq 0\}$

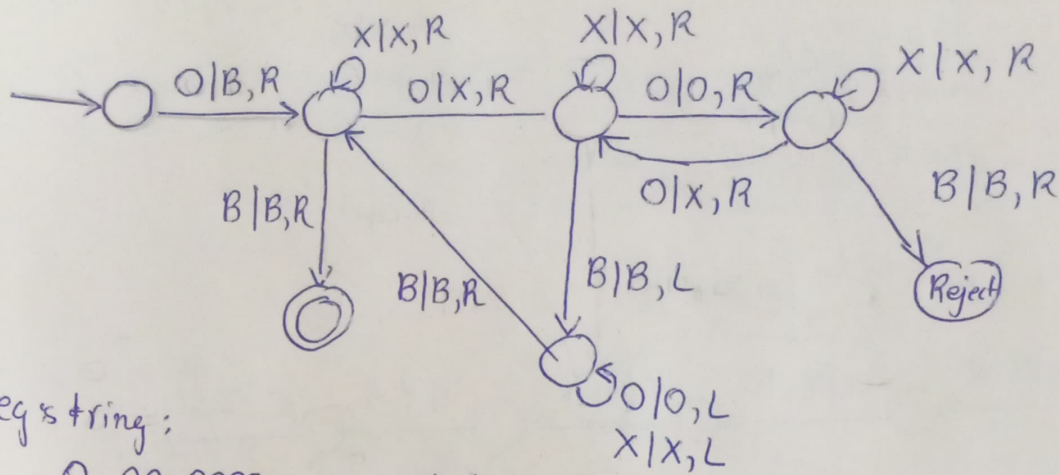


aa bc
 xxy z

aaabcc
 xxy yzz

aaaa bccc
 yxx yzz

iv) $L_3 = \{0^n \mid n \text{ is a power of } 2\}$



eg's string:

0, 00, 0000 ... accepted

000, 00000 ... rejected

test run:

B | 0 | B

B B B
→ B|B, R
accepted state move

B | 0 | 0 | B

B B X B
shift right (X ignore)
X ignore
X ignore
B reached jump accept

B | 0 | 0 | 0 | B

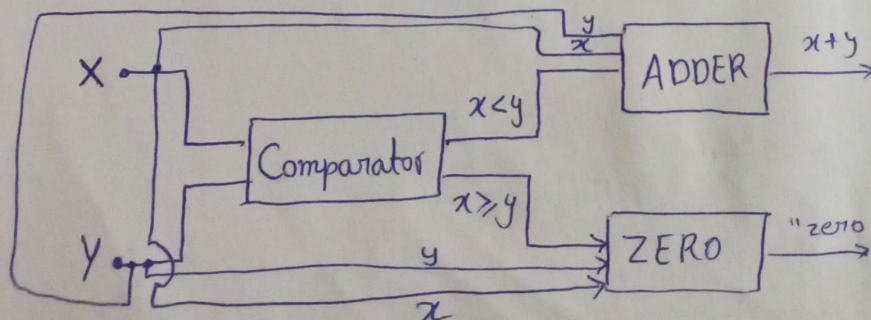
B B X 0 B
B|B, R
jump to rejecter

B | 0 | 0 | 0 | 0 | B

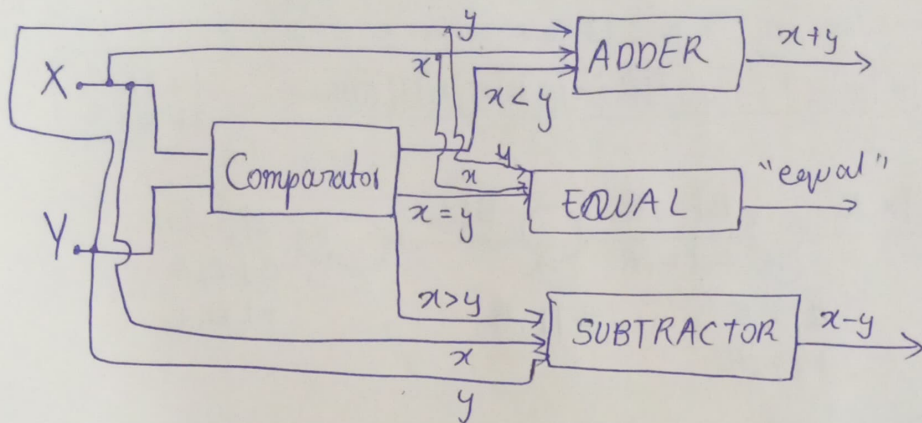
B B X 0 X B
X 0 X
shift left
shift right (X X X B) shift left
shift right (ignore X B) reached accept

v) Turing machine for

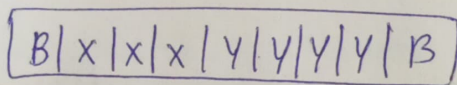
$$f(x, y) = \begin{cases} x+y & x < y \\ \text{"zero"} & x \geq y \end{cases}$$



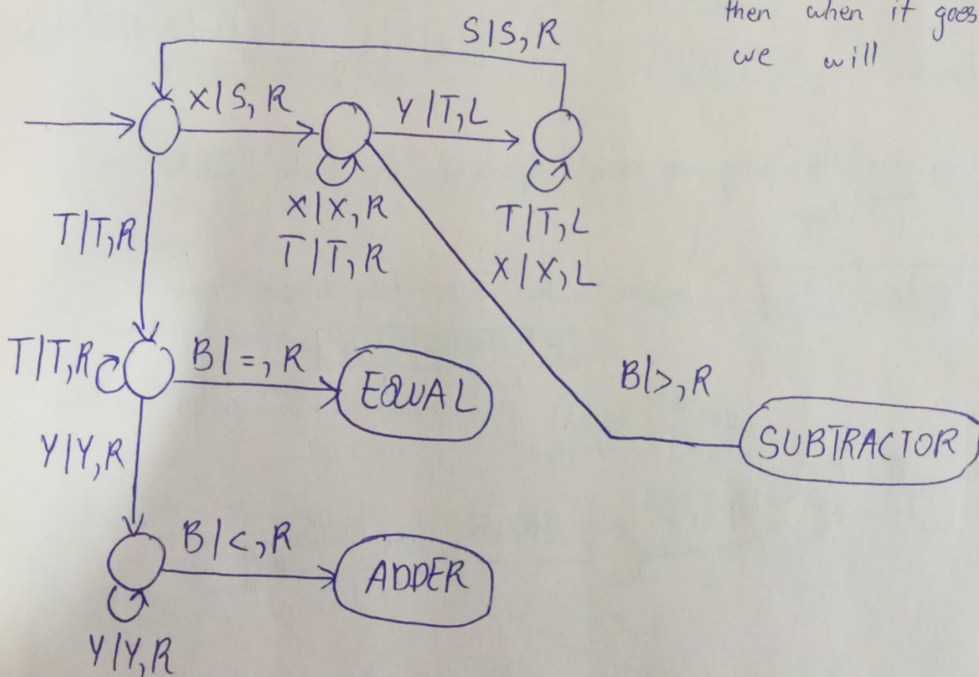
$$vi) f(x,y) = \begin{cases} x+y & x < y \\ x-y & x > y \\ \text{"EQUAL"} & x = y \end{cases}$$



Comparator

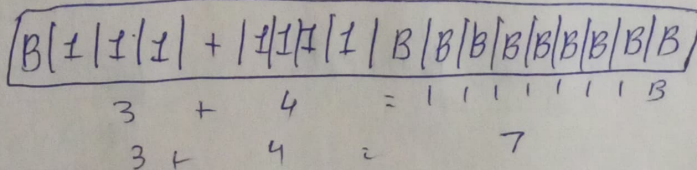


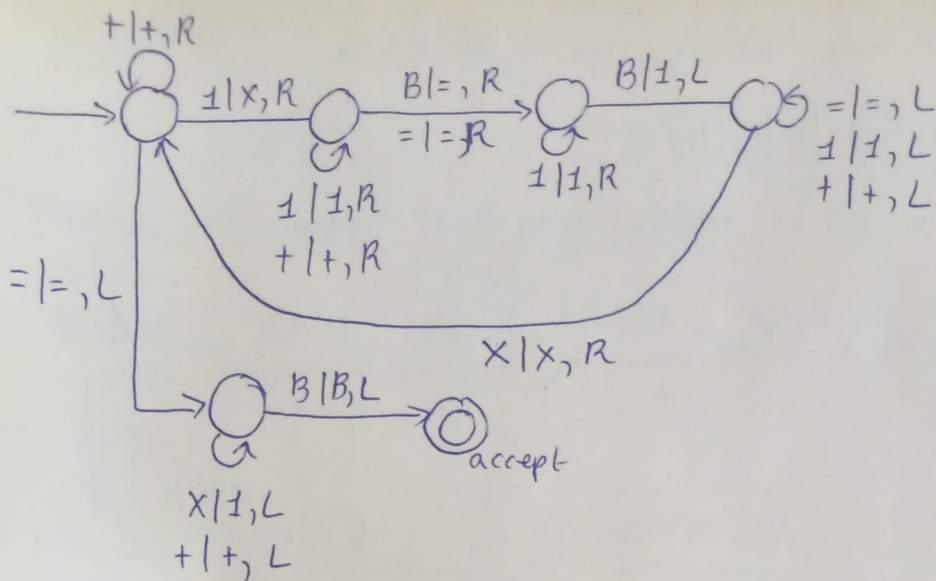
During comparison assuming X and Y are written on tape then when it goes to operation we will



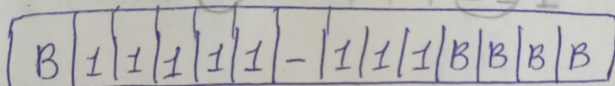
ADDER : $x+y$

as this will be unary addition so x and y will always have alphabet "1" to form their value eg 3 is 111 and so on

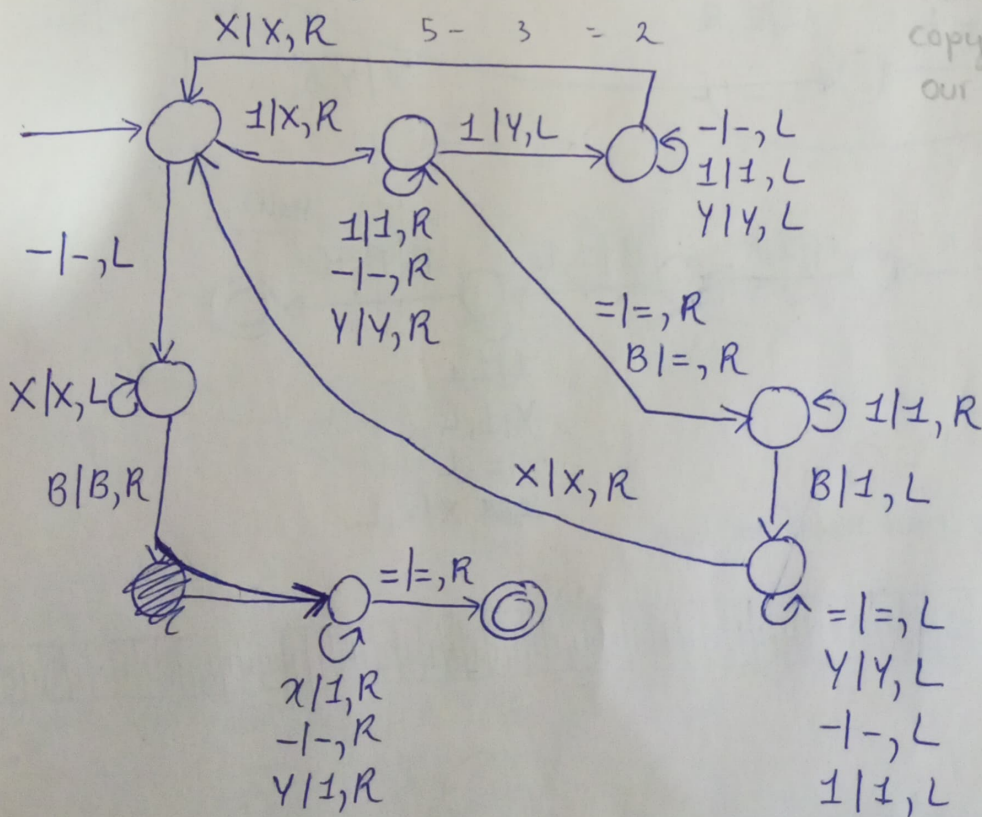




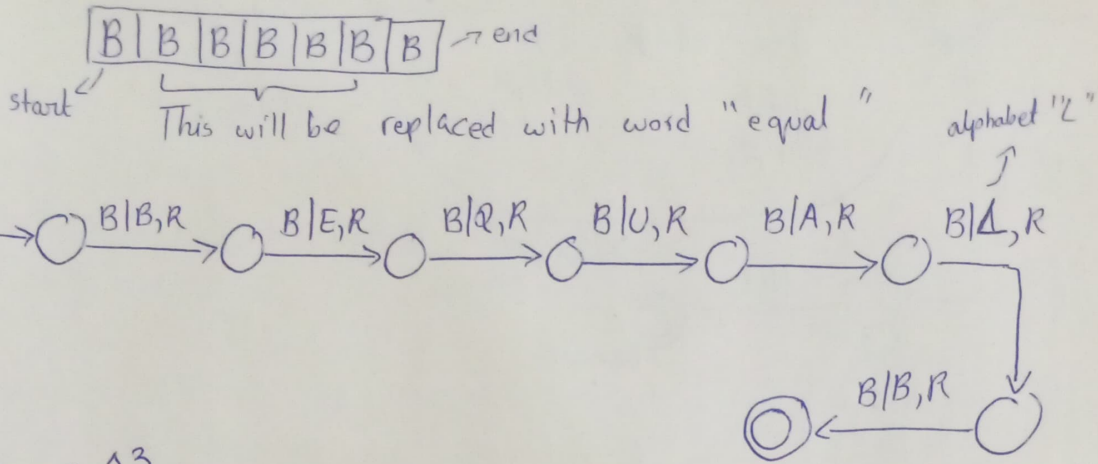
Subtractor: $x - y$ as per comparator result x is always greater than y . so no need to worry of x being equal or less than y . again like adder we will assume x and y values will be denoted as "1"



represent that we now copying 1's for our result.



EQUAL: Assuming initially tape is blank/clean

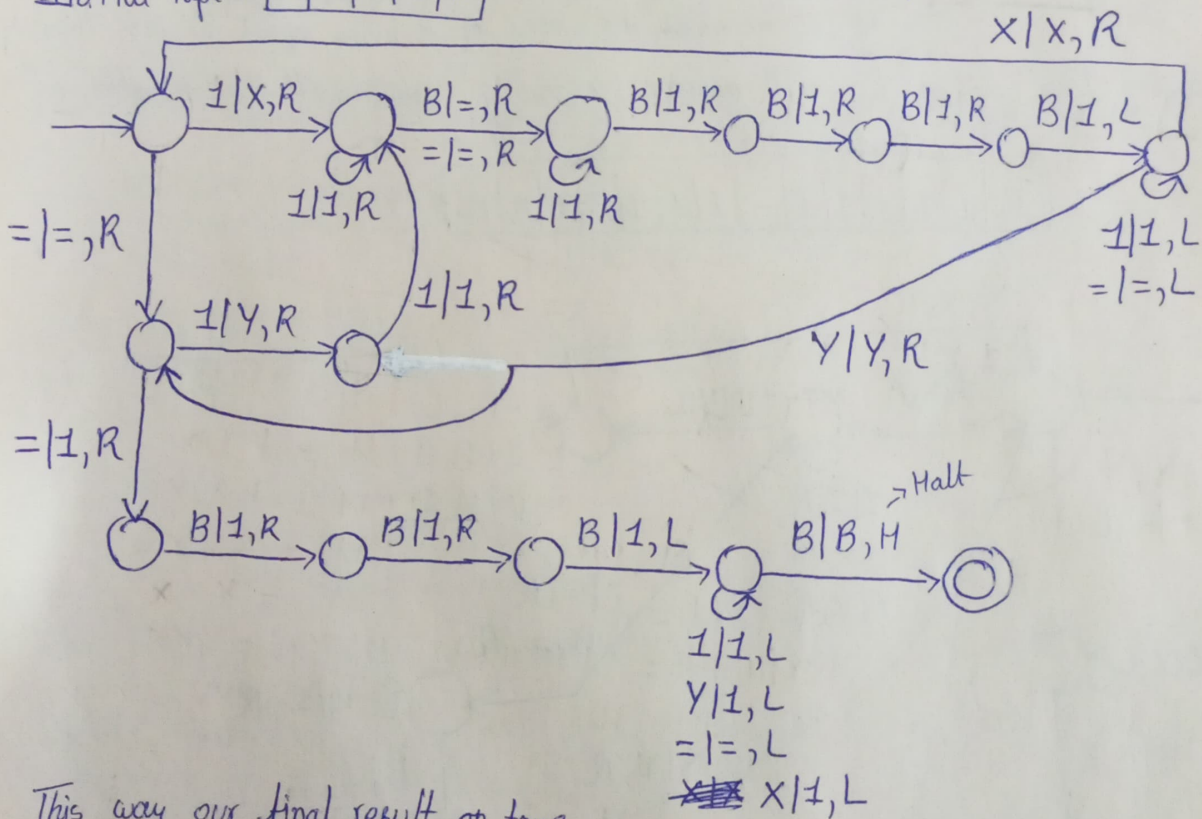


Vii) 4^3

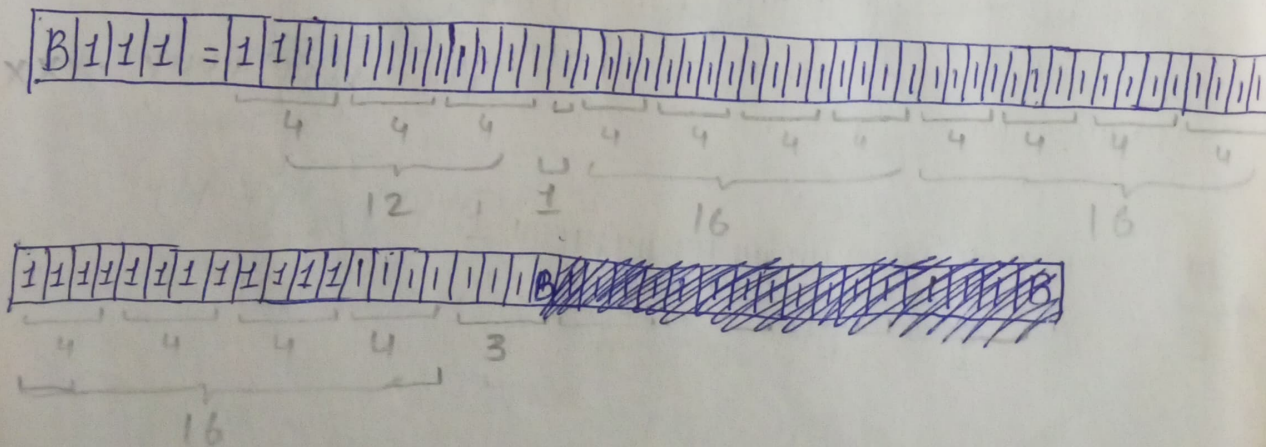
Initial Tape:

B	1	1	1	B
---	---	---	---	---

 these three 1 are the power



This way our final result on tape

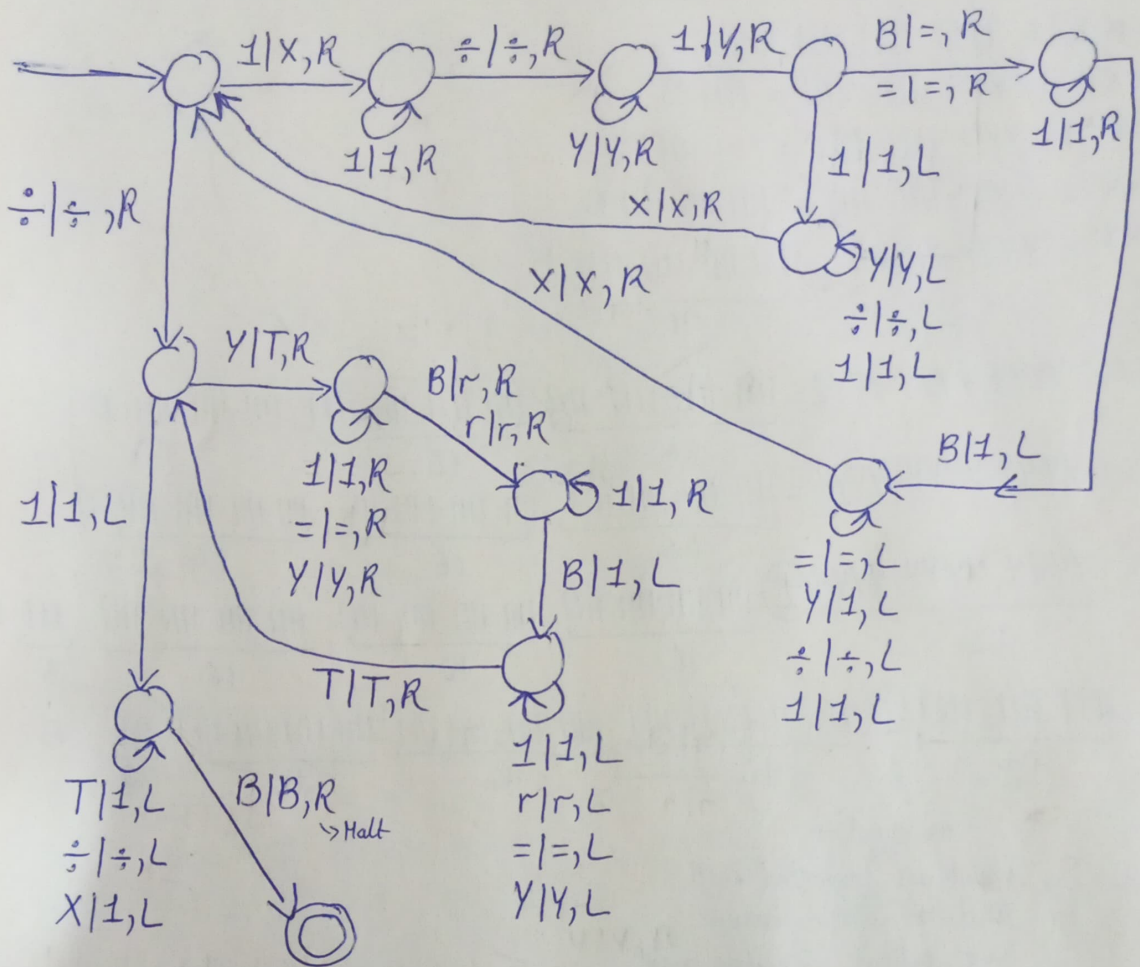
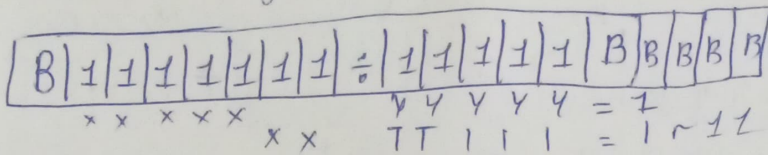


$$12 + 1 + 16 + 16 + 16 + 3 = 4$$

$B \mid 1 \mid 1 \mid 1 \mid B \mid B$

$12 + 1 + 16 + 16 + 16 + 3 = 64$ True proof valid working

VIII) TM divide two unary number



eg-string:

B	1	1	1	1	1	1	1	1	÷	1	1	1	1	1	1	B	B	B	B	B
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

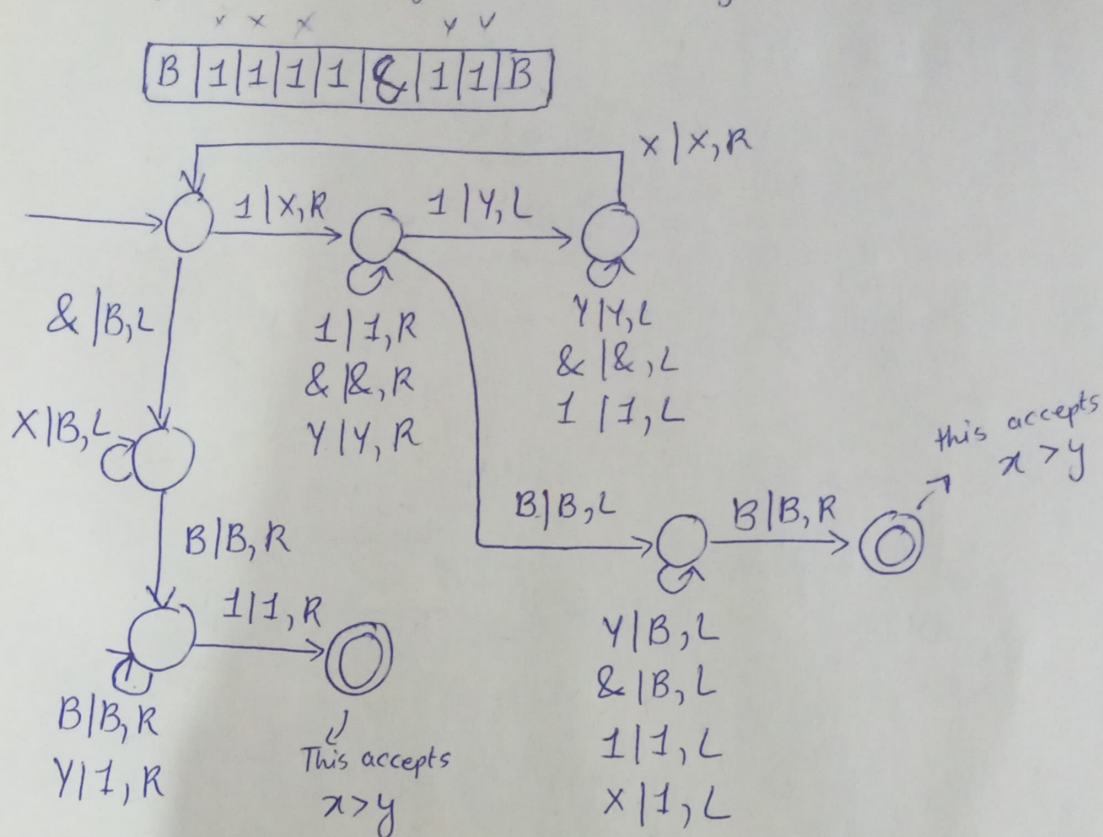
$B\ 11111111\ ÷\ 11111\ B\ B\ B\ B$
 $B\ x\ x\ x\ x\ x\ 11\ ÷\ y\ y\ y\ y\ y\ =\ 1\ B\ B$
 $B\ x\ x\ x\ x\ x\ 11\ ÷\ 11111\ =\ 1\ B\ B$
 $B\ x\ x\ x\ x\ x\ (\otimes)1\ ÷\ y\ 1111\ =\ 1\ B\ B$
 $B\ x\ x\ x\ x\ x\ (\otimes)(\otimes)\ ÷\ y\ y\ 111\ =\ 1\ B\ B$
 $B\ x\ x\ x\ x\ x\ (\otimes)(\otimes)\ ÷\ T\ y\ 111\ =\ 1\ r\ 1\ B$
 $B\ x\ x\ x\ x\ x\ (\otimes)(\otimes)\ ÷\ T\ T\ 111\ =\ 1\ r\ 1\ 1$
 $B\ 11111\ 11\ ÷\ 11111\ =\ 1\ r\ 1\ 1$

$$7 \div 5 = 1r2$$

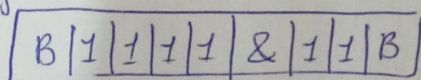
will work for $x > y \Rightarrow Q\ r\ Remainder$
 $x = y \Rightarrow =\ Quotient$
 $x < y \Rightarrow r\ Remainder$

denotes Quotient
denotes remainder

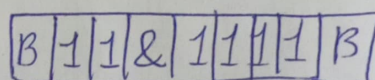
ix) Compare Two unary and output larger.



dry run

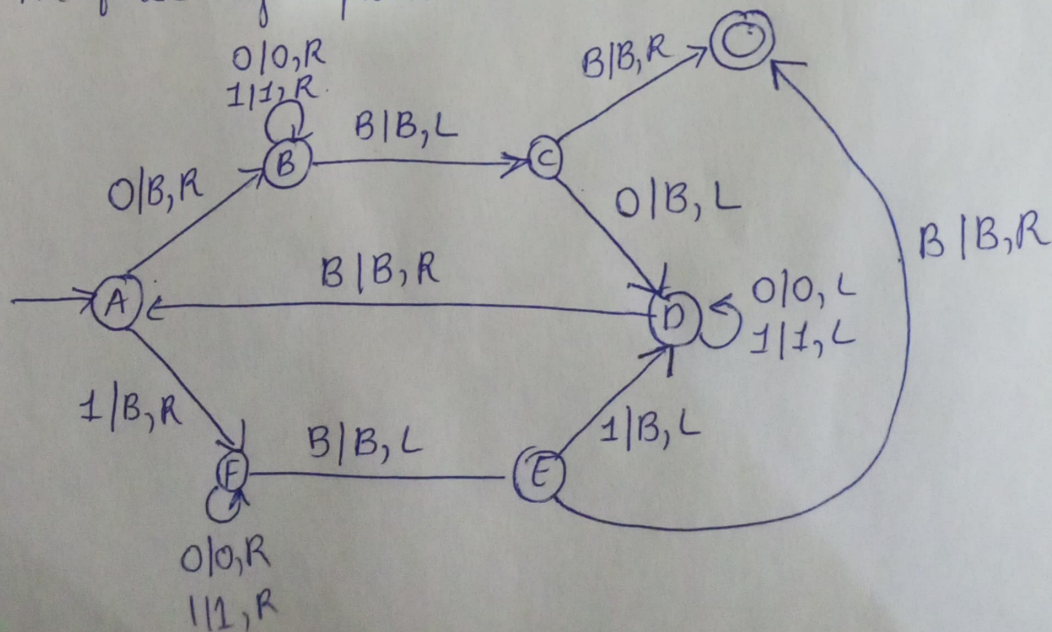


$xx \quad yy$
 $xxx \quad 1 \quad B B B B$
 $B \quad 1 1 1 \quad 1 \quad B B B B \quad \}$ output
 $x > y$



$x \quad 1 \quad \& \quad y \quad 1 1 1 \quad B$
 $xx \quad \& \quad yy \quad 1 1 \quad B$
 $B B B B \quad y \quad y \quad 1 1 \quad B$
 $B B B B \quad 1 1 \quad 1 1 \quad B$ } output accept
 (5th)

X) TM of odd length palindrome



Dry Run

B	0	1	1	1	0	1	0	B
---	---	---	---	---	---	---	---	---

B B 1 0 1 1 1 0 1 B B

0 | B, R one branch

B B B 0 1 1 1 0 B B B

1 | B, R one branch

B B B B 1 1 1 B B B B

0 | B, R one branch

B B B B B 1 B B B B B

1 | B, R one branch

↑
↑
↑
shift
left

← B B

↑ ↑

curr → at right found B so
go to final state.